

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0560

Nickajack Dam 100 kV
January 23, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0560. The Interconnection Customer proposes a 81.5 MW solar photovoltaic facility interconnecting at the Nickajack Dam 100 kV substation in the Illinois area. The request was received on December 31, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 288 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,175,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Carbon Ctr Substation 500-345 kV XFMR, the Blackrock – Shawville Sta 345.0 kV Line loads to 108.4% of its 1,694 MVA rating. Project DFAX: 7.2%.

For the loss of Caldwell – Carbon Center 345.0 kV, the Caldwell – Blackrock 345.0 kV line loads to 106.2% of its 996 MVA rating. Project DFAX: 5.2%.

The Caldwell – Carbon Ctr 345.0 kV is overloaded to 104.1% of its 1,258 MVA emergency rating for the loss of the Blackrock – Shawville Station 345.0 kV. The project contributes 5.1% DFAX (4.1 MW) to this overload.

The Coronet – Ft Humbug 500.0 kV is overloaded to 103.2% of its 2,819 MVA emergency rating for the loss of the Acadia – Montour Sta 500.0 kV. The project contributes 8.4% DFAX (6.9 MW) to this overload.

The Carbon Ctr 500/345kV XFMR is overloaded to 103.1% of its 1,447 MVA emergency rating for the loss of the Blackrock – Shawville Sta 345.0 kV. The project contributes 6.3% DFAX (5.1 MW) to this overload.

The Hilltop – Talenenergy Martins Crk Llc Williamsport 500.0 kV is overloaded to 102.5% of its 2,495 MVA emergency rating for the loss of the Hilltop Substation 500 kV/345 kV TX. The project contributes 9.3% DFAX (7.6 MW) to this overload.

The Hilltop – Strawberry Belt 345.0 kV is overloaded to 101.7% of its 1,615 MVA emergency rating for the loss of the Hilltop – Talenenergy Martins Creek Llc Williamsport 500.0 kV. The project contributes 5.6% DFAX (4.5 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Blackrock – Shawville Sta 345.0 kV Line	Reconductor line	\$16,007,000	1% (\$96,000)
Caldwell – Blackrock 345.0 kV L	Reconductor line	\$22,943,000	1% (\$139,000)
Caldwell – Carbon Ctr 345.0 kV	Reconductor line	\$22,167,000	1% (\$133,000)
Coronet – Ft Humbug 500.0 kV	Reconductor line	\$60,999,000	1% (\$364,000)
Carbon Ctr 500 kV/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$29,988,000	1% (\$181,000)
Hilltop – Talenenergy Martins Crk Llc Williamsport 500.0 kV	Reconductor line	\$32,337,000	1% (\$195,000)
Hilltop – Strawberry Belt 345.0 kV	Reconductor line	\$11,198,000	1% (\$67,000)
	Total allocated to project		\$1,175,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.